

Assignment-1

classmate

Date _____
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Semester: III

Short questions:

① What do we mean by machine learning in the field of artificial intelligence?

→ Machine Learning (ML) is a subfield of artificial intelligence (AI) where algorithms are designed to learn patterns and make predictions or decisions from data, rather than being explicitly programmed for a specific task.

Furthermore it is about creating systems that can improve their performance automatically through experiences (i.e. by being exposed to more data).

② How can learning be defined from a machine's perspective?

→ From a machine's perspective learning is defined by improving its performance on a specific task through experience. This is often formalized by computer scientist Tom Mitchell's definition:

A computer program is said to learn from experience (E) with respect to some class of tasks (T) and a performance measure (P). If its performance at tasks in T, as measured by P, improves with experience E.

- Task (T): The problem the machine needs to solve. (e.g. classifying emails as spam or not spam).
- Experience (E): The data used to train the model. (e.g. a dataset of labeled emails).
- Performance Measure (P): The metric used to evaluate how well the machine performs the task (e.g. the percentage of emails correctly classified).

③ What methods are commonly used to evaluate the performance of machine learning models?

→ The performance of a machine learning model is evaluated using various metrics, which depends on the type of task (e.g. classification or regression).

For classification model:

- These model predict a category (spam or not spam)
- Accuracy: The ratio of correct predictions to the total no. of predictions

$$\therefore \text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total Number of predictions}}$$
- Confusion matrix:
 - A table that summarize the performance of a classification model, showing true positives, true negatives, false positive and false negative
- Precision: Measures the accuracy of positive predictions.

$$\text{Precision} = \frac{\text{True positives}}{\text{True positives} + \text{False positives}}$$

For Regression model:

These models predict a continuous value (e.g. house price).

- Mean Absolute Error (MAE): The average of the absolute difference between the predicted and actual values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Mean Square Error (MSE): The average of the squared difference between the predicted and actual values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Root mean squared Error (RMSE): The squared root of the MSE, which brings the metric back to the original unit of the target variable.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

④ Why is it important to divide data into training and testing sets during model development?

→ Dividing data into training and testing sets is a fundamental practice in ML to prevent overfitting.

• Training set:

- This portion of the data (typically 70-80%) is used to 'teach' the model. The model learns the underlying patterns and relationship from this data.

• Testing set:

- This portion of the data (typically 20-30%) is held back and used to evaluate the model's performance.

after it has been trained. Since the model has never seen this data before, the testing set provides an unbiased estimate of how the model will perform on new, real-world data.

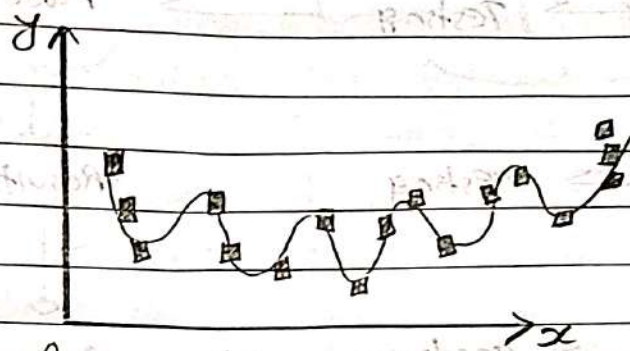


Fig: overfitting

⑤ What does the process of cross-validation involve in model evaluation?

→ Cross-validation is a robust technique used during model evaluation and tuning, particularly for assessing how well a model generalizes to unseen data and preventing overfitting. Instead of a single train-test split, cross-validation involves

- I) Partitioning the data into multiple subsets (folds).
- II) Training the and testing the model multiple times, using different folds for training and testing in each iteration.
- III) Aggregating the results (e.g. averaging performance metrics) to provide a more reliable estimate of the model's performance and to make hyperparameter tuning decisions.

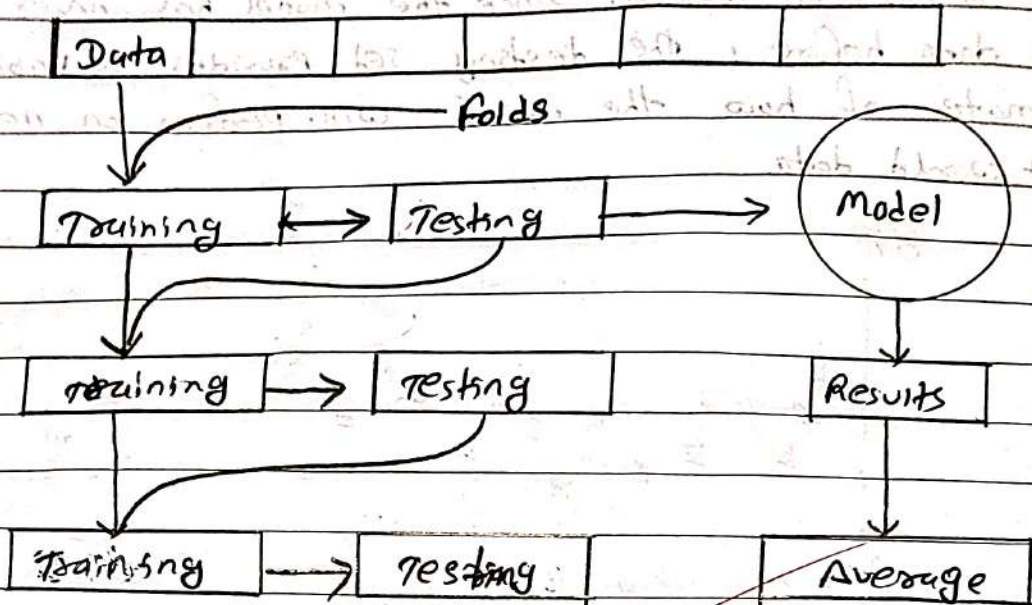


Fig: Cross-validation

⑥ What are feature sets, and how do they influence the learning process of a model?

→ A feature set is the collection of measurable input variables (or attributes) used to train a ML model. The selection and quality of these features are critical as they directly influence a model's ability to learn.

Influence: Models learn by identifying patterns between the input features and the output target. If the features are relevant and informative, the model can easily detect these patterns. If the features are noisy (e.g. to predict credit score using someone's shoe size), the model will struggle to learn and will perform poorly. In short, better features lead to better models.

④ Why is a validation set necessary, and how it is different from test and training sets?

→ A validation set is a subset of the data used to tune the model's hyperparameters and make decisions about the model's architecture. It's different from the training and test sets.

- Training set: used for the model to learn its parameters (e.g. weights)
- Validation set: used to evaluate the model during training to tune hyperparameters (e.g. learning rate, no. of layers) and prevent overfitting. It acts as unbiased check during the model development phase.
- Test set: used only once at the very end to provide a final, unbiased assessment of the fully trained and tuned model's performance on unseen data.

Using a validation set prevents 'leaking' information from the test set into model tuning process, which would make the final evaluation overly optimistic.

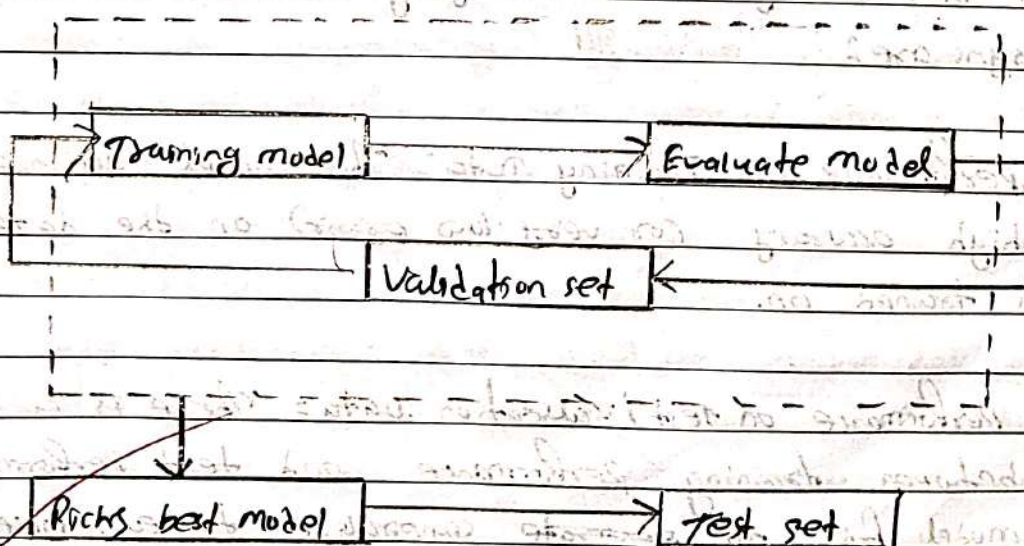


Fig: Training data / validation / test

⑧ What is the purpose of splitting a dataset into multiple subsets before training a model.

→ Splitting a dataset into multiple subsets (training, validation and testing) is essential for developing a reliable and generalizable model. The primary purpose is to simulate how the model will perform on new, real-world data and to avoid overfitting. This separation allows you to:

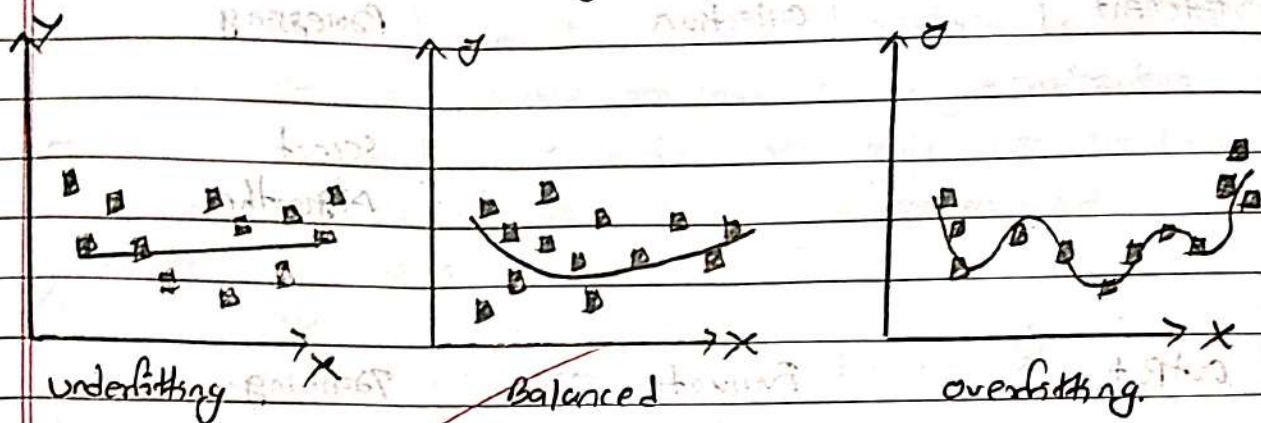
- i) Train the model on one portion of data.
- ii) Tune and select the best model using a separate validation portion.
- iii) Evaluate the model its final performance honestly on a completely unseen test portion.

⑨ What are the key signs that a model is overfitting the training data?

→ A model is overfitting when it learns the training data too well, including its noise and random fluctuations, instead of the general underlying pattern. The key signs are:

- High performance on Training Data: The model achieves very high accuracy (or very low error) on the data it was trained on.
- Poor performance on test/validation data: There is a large gap between training performance and test performance. The model fails to generalize to new, unseen data.

- Excessive model complexity: The model might be too complex for the amount of data available (eg a deep neural network for a ~~big~~ simple problem).



⑩ How does a machine learning model learn from patterns in data?

→ A machine learning model learns by mathematically optimizing its internal parameters to map input data to correct outputs. The process generally involves these steps:

- i) Prediction: The model takes an input from the training data and makes a prediction.
- ii) Error calculation: It compares its prediction to the actual known outcome and calculates an error or loss.
- iii) Optimization: An optimization algorithm uses this error to slightly adjust the model's internal parameter in a direction that would reduce the error.
- iv) Iteration: This process is repeated thousands or millions of times with all the data in the training set. With each iteration, the model's parameters get progressively better at capturing the patterns that connect the features to the correct outcomes.

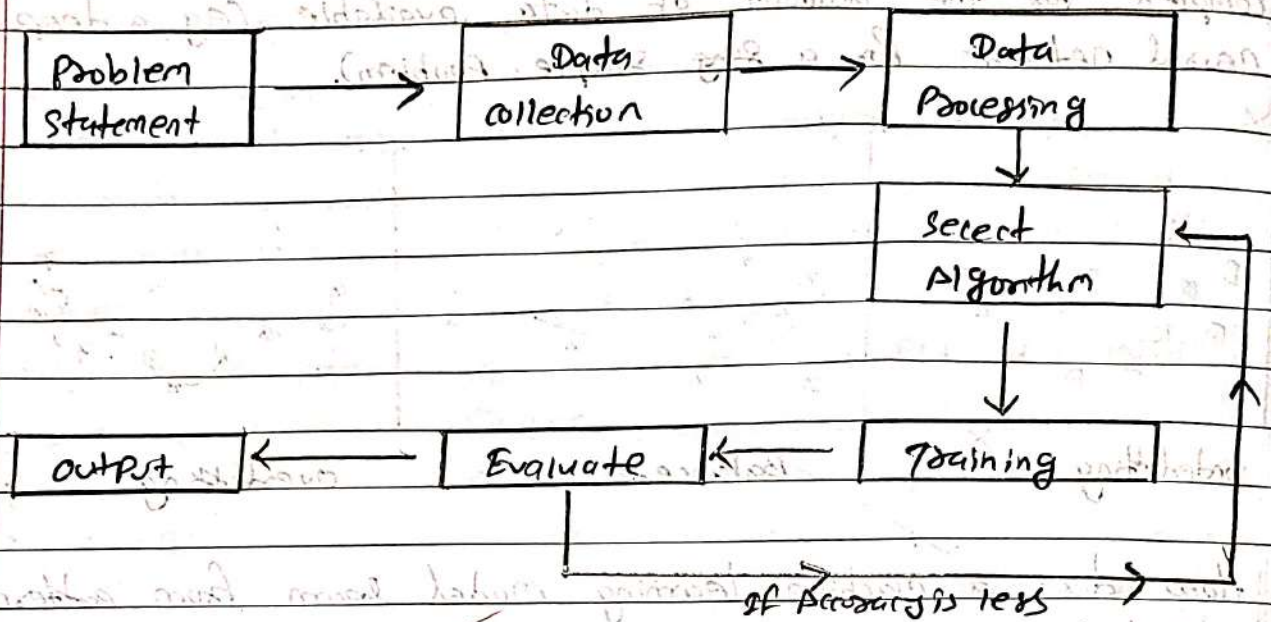


Fig: Machine learning model

(I) In machine learning, what constitutes a dataset and what elements does it typically include?

→ In ML, a dataset is a structured collection of examples, typically organised in a table format.

- **Samples (or Instances):** The rows of the tables. Each row represents a single observation or data point.
- **Features:** The columns of the table. Each column represents a characteristic or attribute of the sample.
- **Target (or Label):** A special column containing the outcome value that the model aims to predict. This is the 'answer' that the model learns from in supervised learning.

(12) How does cross-validation improve the generalization ability of model?

→ Cross-validation improves a model's development process by providing a more reliable estimate of its generalization ability. By training and testing the model on multiple different subsets of the data, it averages out the performance metrics.

- This gives a more stable and less biased measure of how the model will likely perform on unseen data compared to a single train-test split. This better estimate helps developers select the best-performing model and hyperparameters, which indirectly leads to a final model that generalizes better.

(13) What are the main evaluation metrics used to measure a model's accuracy and reliability?

→ The main metrics to measure a model's accuracy and reliability depends on the task.

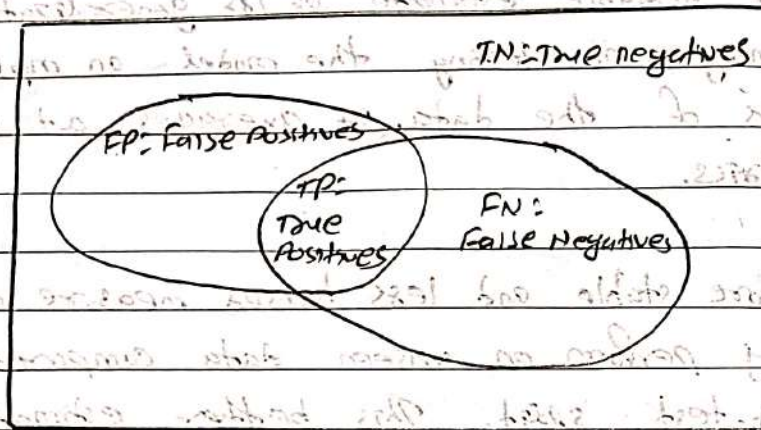
1) For classification (Predicting categories):

- Accuracy: overall correctness
- Precision: Accuracy of positive predictions
- Recall: Ability to find all actual positives
- F1-score: A single score that balances precision and recall

2) For Regression (Predicting numerical values):

- Mean Absolute Error (MAE): Average absolute prediction error.

- Mean squared error: (MSE): Average squared error (penalize large errors more)
- Root mean squared error: square root of MSE, expressed in the same units as target.



$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall (r)} = \frac{TP}{TP + FN}$$

$$\text{F-measure (F)} = \frac{2}{\frac{1}{p} + \frac{1}{r}}$$

$$= \frac{2pr}{p+r}$$

$$\text{Accuracy (acc)} = \frac{TP + TN}{TP + TN + FP + FN}$$

④ How do training and testing datasets differ in their roles and usage?

→ The roles of training and testing datasets are distinct and crucial for proper model development.

- Training dataset: This is the data used to teach the model. The model iterates over this data to learn the underlying patterns by adjusting its internal parameters. It is the "study material".
- Testing Dataset: This data is kept separate and is used to only at the end to evaluate the final, trained model. It acts as a "final exam" to get an unbiased measure of the model's performance on completely new data.

Q1) What Datasets and we can explain it?

→ A classic example of dataset used in ML is the Iris flower dataset.

- A dataset is a collection of organized data - like a table or spreadsheet. Each row is an item, each column is a feature.

Types of Datasets:

- Structured - clean, like Excel files.
- Unstructured - messy, like tweets or images.
- Semi-structured - in-between like JSON.

- Dataset powers AI, machine learning, and analytics. No data = no smart tech.

Long Questions:

Q1) Explain the history and evolving of Machine Learning with examples.

→ ML journey begins long before computers. Its roots are in statistics and pattern recognition, with foundational ideas like Bayes' Theorem (1700s), and linear regression laying the mathematical groundwork.

The term 'Machine Learning' was coined in 1959 by Arthur Samuel, who created a checkers playing program that learned from its own games to improve its performance.

This field was revitalized in the 1980s with the rediscovery of backpropagation, an algorithm that

made it practical to train more complex neural networks. The 1990s saw the rise of powerful methods like support vector machine (SVMs). The modern era of deep learning exploded in the 2010s, driven by three key facts:

- 1) Big data: The availability of massive datasets to train on.
- 2) Powerful Hardware: The use of GPUs for parallel processing.
- 3) Algorithmic Advances: Innovations like AlexNet model in 2012, which revolutionized image recognition.

② What is machine learning, how does it differ from traditional rule-based programming, and why is it considered a core part of AI?

→ The core difference lies in the programming paradigm:

- Traditional programming
 - A developer writes explicit, step-by-step rules. The computer takes this program and input data to produce an answer. The logic is handcrafted.
- Machine Learning
 - A developer provides a model, input data and the corresponding answers. The system uses this to learn the underlying rules that connect the inputs to the answers. The logic is inferred from data.

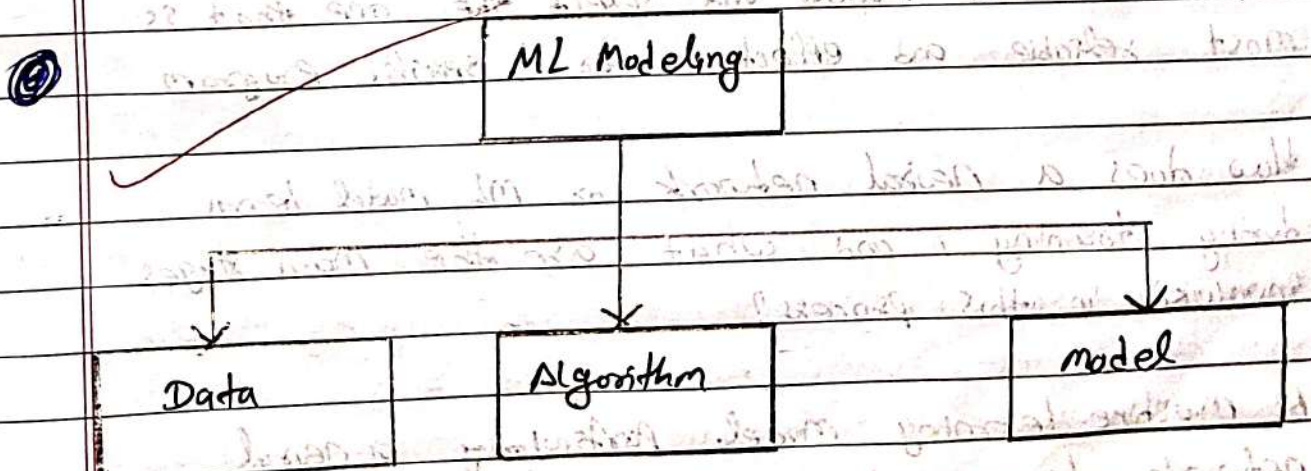
ML is considered a core part of AI because learning from experience is a fundamental characteristics of intelligence.

③ How do we define "learning" in the context of ML systems, and what are the key components that enables a model to learn from data?

→ In ML 'learning' is defined as a process of improving performance on a task through experience. A system learns if its ability to perform a task gets better as it's exposed to more data.

The key components that enables a model to learn are:-

- i) Data : The 'experience' the model uses to find patterns. This includes input features and, in supervised learning, the correct output labels.
- ii) model : A mathematical framework with adjustable parameters that tries to represent the relationship between the data's features and its labels.
- iii) Loss functions : A function that measures the model's error by comparing its prediction to the actual label.



④ What are the different evaluation methods used in machine learning, and how do they help in assessing the quality and performance of a model?

→ Evaluation methods are used to assess a model's quality and performance on unseen data, ensuring it generalizes well and isn't just 'memorizing' the training set. The choice of methods depends on the task.

For classification tasks:

- Accuracy
- Precision
- Recall
- F1-score

For Regression tasks:

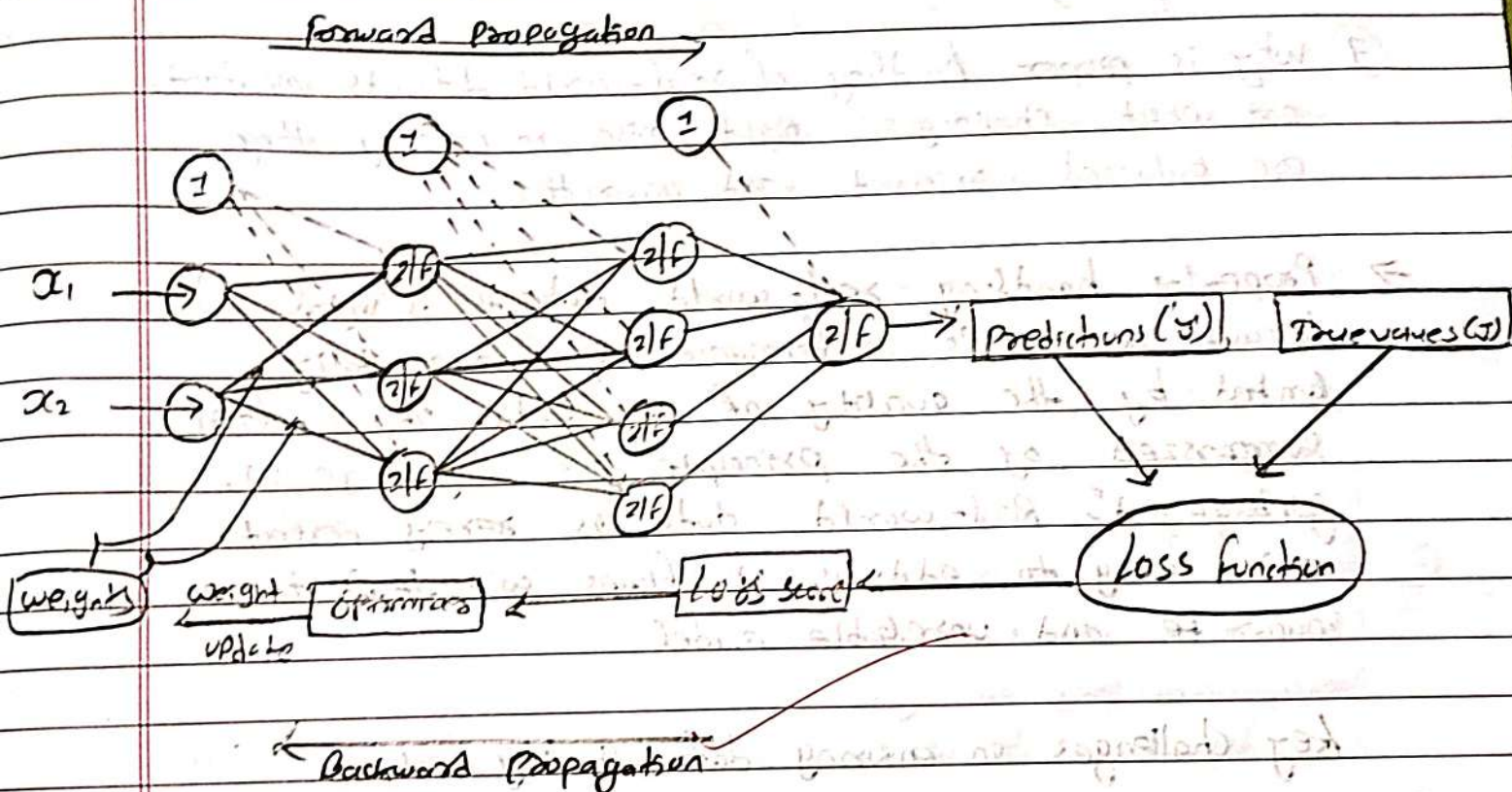
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

These metrics help us quantify a model's performance, compare different models and select the one that is most reliable and effective for a specific program.

⑤ How does a neural network or ML model learn during training, and what are the main stages involved in this process?

- A machine learning model, particularly a neural network, learns during training through an iterative optimization process. This process, often called the training loop, involves several main stages:

- Initialization
- Forward Pass
- Loss Calculation
- Backward Pass
- Parameter update



⑥ What exactly are datasets in machine learning, and how should they be collected, cleaned, and structured for effective model training?

→ In ML, a dataset is a structured collection of examples used to train and evaluate a model. It's typically organized as a table where rows represent individual samples (customers, a house, etc.) and columns represent features. Often, one column is the target or label.

Preparing a dataset for effective model training involves three key steps:

- i) Collection
- ii) cleaning
- iii) Structuring

⑦ Why is proper handling of real-world datasets important, and what challenges might arise in ensuring they are balanced, relevant, and accurate?

→ Properly handling real-world datasets is vital because a model's performance is fundamentally limited by the quality of its data. This is often summarized by the principle of "garbage in, garbage out". Real-world data is rarely perfect, and failing to address its flaws will lead to an inaccurate and unreliable model.

Key challenges in ensuring data quality includes:

- Imbalanced data
- Relevance
- Accuracy
- Errors

⑧ What are features sets in machine learning, and how does the selection, transformation, or extraction of features impact model performance?

→ A feature set is the group of input variables selected from the dataset to train a model. The

Process of selecting and modifying these features, known as Feature engineering, has a massive impact on a model performance.

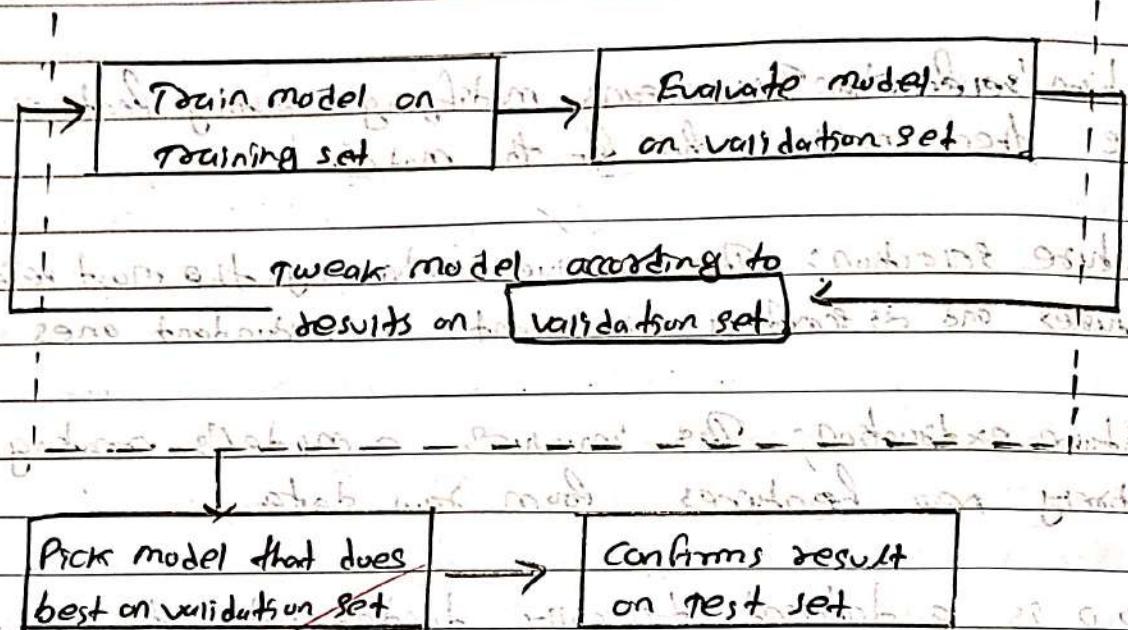
- Feature transformation: This means modifying existing features to make them more useful for the model.
- Feature selection: This involves choosing the most relevant features and discarding irrelevant or redundant ones.
- Feature extraction: This involves a model's creating entirely new features from raw data.

⑨ How is a dataset typically divided into training, validation, and testing sets and what specific roles does each set play in model development and evaluation?

→ A dataset is typically divided into three distinct subsets to ensure robust model development and evaluation. The split is usually done randomly to maintain a similar data distribution across all subsets. A common split ratio is 70% for training, 15% for validation, and 15% for testing.

The steps are as:

- i) Training set
- ii) Validation set
- iii) Testing set

Diagram:

⑩ What is cross-validation, how is it performed, and why is it considered a reliable technique for model evaluation and tuning?

→ cross-validation is a powerful and reliable evaluation technique that provides a more stable estimate of model's performance than a single train-validation split. The most common method is k -fold cross-validation.

Here's how it's performed (ex with $k=5$).

1) split: The dataset is randomly shuffled and split into 5 equal-sized folds.

- ii) Iterate: The process runs for 5 iterations. In each iteration:
- one fold is held out as the test set.
 - The remaining 4 folds are combined to form the training set.
 - The model is trained and then evaluated on the test fold.
- iii) Average: The final performance metric (e.g. accuracy) is the average of the scores from all 5 iterations.

It's considered a reliable technique because every data point gets to be in a test set exactly once. This minimizes the risk that the evaluation is skewed by a single "lucky" or "unlucky" split, giving a much better sense of how the model will generalize to unseen data.

28/7/25

Assignment:- 2

Date: SPS
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Semester: III

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Submission date: 16/09/2025

Short questions:-

① Define Machine Learning?

→ Machine Learning (ML) is a subset of Artificial Intelligence (AI) that allows computers to learn from data and make decisions without being explicitly programmed. Essentially, ML algorithms identify patterns in data and use those patterns to predict or classify new, similar data. This ability to learn from data and improve over time is what makes ML so powerful.

② List any three real-life applications of Machine Learning?

→ The any three real-life applications of ML are as follows:-

1) Recommendation Engine: Services like Netflix, Spotify use ML to suggest movies, shows or songs based on your past viewing and listening habits.

i) NLP: Voice assistants like Siri and Google Assistant responding to commands, translation tool like Google Translate translating between languages and chatbots providing customer support.

iii) Autonomous vehicles: Self-driving cars navigating streets and making driving decisions, and ride-sharing apps like Uber optimizing routes and pricing.

③ What are the main processes involved in Machine Learning?

→ The ML lifecycle is a systematic process of developing and deploying effective ML models, consisting of several stages

- Problem Definition
- Data collection and preparation
- Model selection and Engineering
- Training and validation
- Evaluation and testing
- Deployment and Integration
- Monitoring and Maintenance

4) Differentiate between supervised and unsupervised Learning?

→	Supervised Learning	Unsupervised Learning
i)	uses labeled data (data with known outcomes)	uses unlabeled data (data with no predefined outcomes).
ii)	To predict an outcome based on the training data.	To discover hidden patterns in the data.
iii)	It is learn with a teacher who provides correct answer	Learning on your own by finding similarities or anomalies.

⑤ Give one example of each of classification and regression problems?

→ Classification: Predicting whether an email is spam or not spam. The output is a discrete category.

→ Regression: Predicting the price of a house based on its features (like size, location, and number of bedrooms). The output is a continuous value.

⑥ What is Reinforcement Learning?

→ Reinforcement Learning (RL) is a type of machine learning where an agent learns to make decisions by performing actions in an environment to maximize a cumulative reward. The agent learns through trial and error, much like training a pet with treats for good behaviour. It's not given explicit instructions, but rather a goal to achieve.

⑦ Name two algorithms used in Supervised Learning?

→ The any two algorithms used in supervised learning are as:-

1) Linear regression: Predicts outcomes by fitting the best straight line through data.

2) Decision Tree: Predicts by splitting data into branches based on rules.

⑧ Name two algorithms used in unsupervised learning?

→ The only two algorithms used in unsupervised learning are:

- 1) k-means clustering: Groups data into k clusters based on similarity.
- 2) PCA (Principal Component Analysis): Reduces data dimensions by keeping the most important features.

⑨ Define training dataset and test dataset?

→ Training dataset

- This is the portion of the data used to train the ML model. The model learns the underlying patterns and relationships from this dataset.
- It's like the set of practice problems you study for an exam.

Test dataset

- This is a separate portion of the data that the model has never seen before. It's used to evaluate the model's performance and check how well it generalizes to new, unseen data. It's like the final exam itself.

⑩ What is the role of a model in Machine Learning?

→ In Machine Learning, a model is the mathematical representation of a real-world process that is the output of the training algorithm.

Essentially, it's the "brain" that has learned from the training data. Its role is to take new input data and make predictions or decisions based on the patterns it has learned.

⑪ State any two advantages of Machine Learning?

→ The advantages are:

i) Automation: ML can automate repetitive and complex tasks, such as data entry, analysis, and decision-making, increasing efficiency and reducing human error.

ii) Pattern Identification: It excels at identifying trends and patterns in large and complex datasets that might be impossible for a human to spot.

⑫ Mention one real-life example of Reinforcement Learning?

→ A great example is self-driving cars. The car's AI acts as the agent, the roads and traffic are the environment and it receives positive rewards for actions like staying in its lane and reaching the destination safely and negative rewards for actions like collisions or breaking traffic laws.

(13) What is the purpose of features selection in machine learning?

→ The purpose of features selection in ML is to choose the most relevant input variables (features) for creating a model. This helps to:

- (a) Improve the model's predictive accuracy.
- (b) Reduce overfitting (when a model performs well on training data but poorly on new data).
- (c) Decrease the computational time and resources needed for training.

(14) List two common challenges in implementing machine learning?

→ The two common challenges in implementing machine learning are as follows:

- i) Data quality and quantity
 - ML models require large amounts of high-quality, relevant data. Insufficient or dirty data leads to poor performance.

- ii) overfitting:
 - The model may learn the training data too well, including its noise and outliers, causing it to fail when exposed to new, unseen data.

⑤ Give one example of machine learning in healthcare.

→ One key example is medical image analysis. ML algorithms are trained to analyze medical scans like X-rays, CT scans, and MRIs to detect diseases such as concave tumors, diabetic retinopathy in eye scans, or signs of Alzheimer's in brain scans, often with accuracy comparable to or exceeding human radiologists.

Long Questions:-

① Explain the various processes involved in machine learning with a neat diagram.

→ Machine learning process involves several key stages that transform raw data into a functional model that can make predictions or decisions.

I) Problem definition

- clearly define the problem that needs to be solved and the objectives to be achieved.

II) Data collection and prediction

→ Gather, cleaning, and labelling data which is crucial for building and training an accurate model.

III) Model selection and Engineering:

- Choosing the right algorithms and building the model architecture based on the problem and data type.

IV) Training and Validation

- Training the model with the prepared data and validating its performance on unseen data to avoid overfitting and underfitting

V) Evaluating and testing:

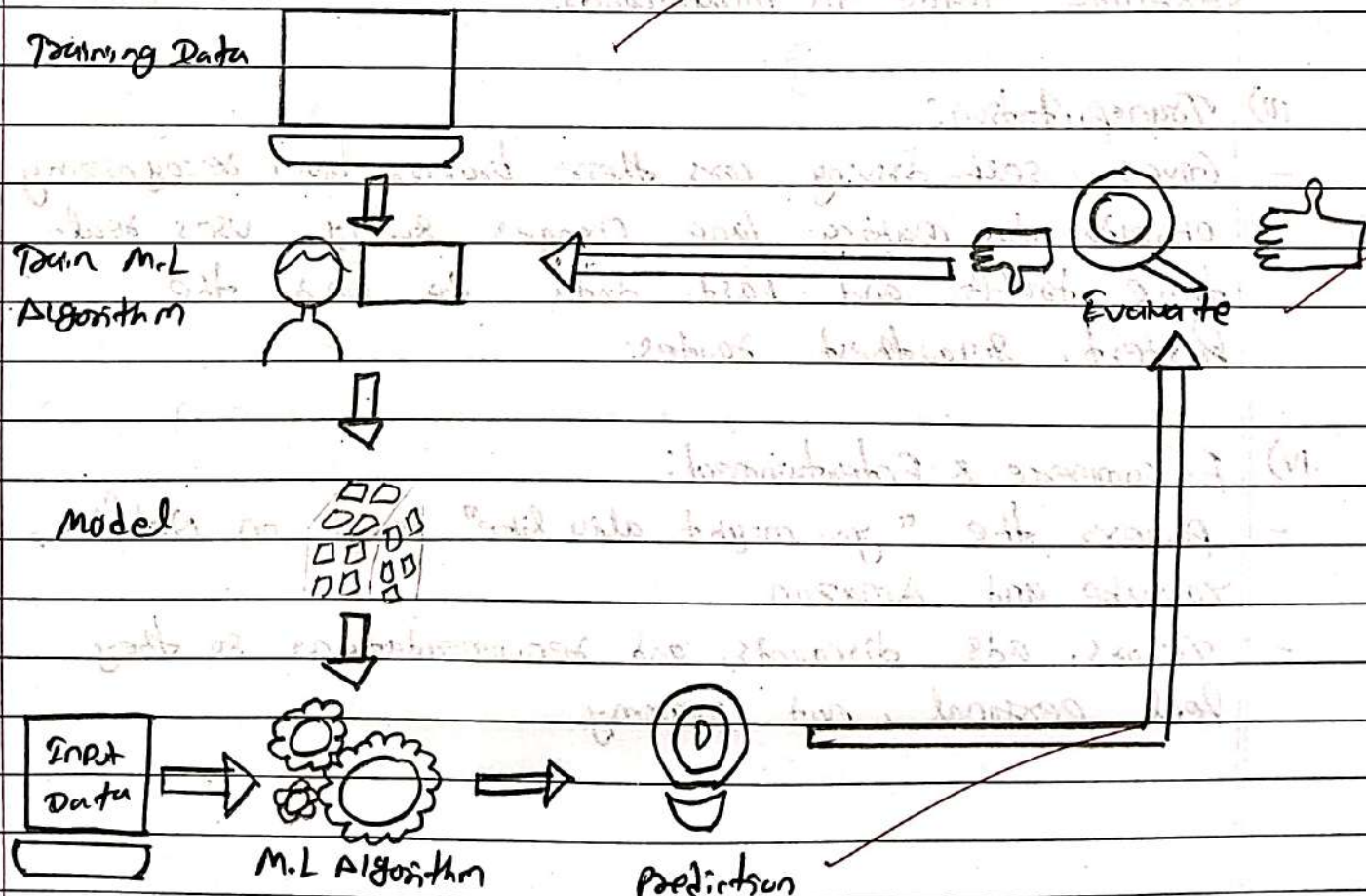
- Rigorously evaluating the model's performance on test data using relevant metrics and making improvements

VI) Deployment and Integration:

- Deploying the model into a live environment, making its predictions accessible through APIs, web apps.

VII) Monitoring and maintenance:

- continuously monitoring the model's performance and retraining it with new data to ensure continued effectiveness.



② Discuss real-life applications of Machine Learning in different domains?

→ ML is transforming numerous industries, here are some key applications.

i) Healthcare:

- Scans - MRIs, CTs, and X-rays to catch tumors or diseases earlier than most doctors can. speed up drug discovery by predicting which compound might actually work before they hit the lab.

ii) Finance

- watches every transactions in real time to catch fraud before it drains your accounts. Runs algorithmic trading bots that can predict market shifts and execute trade in milliseconds.

iii) Transportation:

- Gives self-driving cars their brains - from recognizing objects to making lane changes safely. uses real-time traffic and past data to find the fastest, smoothest routes.

iv) E-commerce & Entertainment:

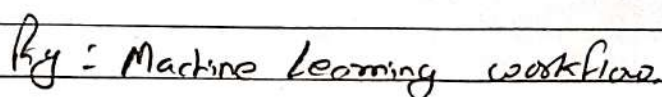
- Powers the "you might also like" magic on Netflix, Youtube and Amazon.
- Tailors ads discounts, and recommendations so they feel personal, not spammy.

③ compare Supervised, unsupervised and Reinforcement Learnings with examples.

→ Supervised Learning	unsupervised Learning	Reinforcement Learning
i) Labeled data (input-output pairs)	unlabeled data (input only)	No predefined data: interacts with environment.
ii) To predict an output based on input data	To discover hidden patterns or structures in data.	To learn the best sequence of action to maximize a reward.
iii) Direct feedback from the labeled data.	No direct feedback; learns inherent structures.	Feedback through rewards or penalties from action.
iv) Example Spam detection: The model is trained on emails labeled as "spam" or "not spam" to classify new emails.	Example Customer segmentation: Grouping customers with similar purchasing habits for targeted marketing.	Example: Game playing: An AI agent learns to play a game like chess by getting rewards for good moves and penalties for bad ones.

→ A typical ML project follows a structured workflow from conception to deployment.

- i) Define - Problem: know what you're solving & set goal.
- ii) collect data: Grab all the relevant info
- iii) Clean & Prep: fix, format and split data
- iv) Explore - spot pattern & insights
- v) Pick model - Choose the right algorithm
- vi) train model - Train it with your data
- vii) Evaluate - test accuracy & performance
- viii) Deploy - Put it into real-world use
- ix) Monitor - keep it fresh & update when needed.



⑤ Explain Supervised Learning - Discuss its types with suitable examples?

→ It is a type of machine learning where the algorithm learns from data that is labeled. This means for each data input, there is a corresponding correct output. The goal is to learn a mapping function that can predict the output variables (y) from the input data (x). The formula is often represented as $y = f(x)$.

Two types are

i) Classification: The output variable is a category or a class. The model predicts a discrete label.

Ex: Image Recognition: A model is trained on a dataset of thousand image of cat and dogs after training the model can predict whether it's cat or dog.

ii) Regression: The output variable is a continuous, real value. The model predicts a quantity.

Ex: House Price Prediction. A model is trained on a dataset of houses where the features include square footage, no of bedrooms and location, and the label is the price. The trained model can then predict the price of a new house based on the features.

⑥ Explain unsupervised Learning. Discuss clustering and dimensionality reduction techniques?

→ Unsupervised Learning is used when you have data with no labels. The goal is to allow the model to explore the data on its own to find meaningful structures or patterns. It's about discovering the underlying distribution of the data.

Techniques

i) Clustering

- This technique involves grouping a set of data points in such a way that points in the same group are more similar to each other than to those in other groups.

ex:

k-means clustering: This algorithm partitions data into k distinct, non-overlapping clusters.

ii) Dimensionality Reduction:

- This technique is used to reduce the number of features in a dataset while retaining as much important information as possible. It helps in simplifying models, reducing storage space, and speeding up training.

ex:

Principal Component Analysis (PCA). It transforms the original features into a new set of uncorrelated features called principal components.

⑦ Explain the concept of Reinforcement Learning with a suitable real-life example?

→ Reinforcement Learning is a goal-oriented learning paradigm where an agent learns to make decisions by performing actions in an environment to maximize a cumulative reward. Unlike supervised learning, there's no correct labeled data. The agent learns from the consequences of its actions through trial and error.

Example: Training a Robot vacuum cleaner

Imagine a new robot vacuum cleaner learning to efficiently clean a room

- Agent: The robot vacuum itself.
- Environment: The room, including furniture, carpet, and obstacles.
- State: The robot's current location, battery level, and sensor reading about nearby objects.
- Action: move forward, turn left, turn right, start suction
- Rewards according to action good and bad

⑧ Describe any three Machine Learning applications in business and industry

→ ML provides a competitive edge to business across various sectors

i) customer churn prediction

- In telecom banking, and subscriptions, ML analyzes customer behaviour, contracts, and service history to predict who's likely to leave, allowing proactive retention with targeted offers or better service.

ii) Predictive maintenance in manufacturing:

- unexpected equipment failure can halt production lines, leading to significant financial losses. ML processes sensor data like vibration and temperature to detect any signs of equipment failure, enabling timely maintenance that reduces downtime and costs.

iii) Dynamic Pricing

- In e-commerce, travel, and ride-sharing, ML adjusts prices in real time based on demand, competition, weather and events, helping boost revenue and attract customers.

⑨ Discuss the importance of data preprocessing in the machine learning process?

→ Data preprocessing is the crucial step of cleaning and organizing raw data to make it suitable for a machine learning model. Its importance is often summarized by the principle "Garbage In, Garbage Out". A sophisticated algorithm will produce poor results if fed low-quality data.

Some Important Points:

- Boosts Accuracy & Performance: Clean, structured data helps models learn patterns effectively, while noisy data leads to poor predictions.
- Handles missing & Inconsistent Data: Techniques like imputation fill gaps, and standardization unifies formats (e.g. USA vs "United States").
- Makes Data Model-Ready: Converts categorical values into numerical form so algorithms can process them.
- Prevents Bias in Results: Scaling features ensures algorithms don't favor variables with large numeric ranges.

⑩ Explain the advantages, disadvantages and limitations of Machine learning?

→ Advantages

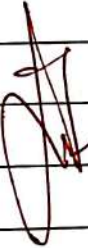
- i) Automation of Tasks - ML reduces the need for manual intervention by automating decision-making and predictions.
- ii) Improved Accuracy over Time - models learn from more data, often becoming more accurate as they're retrained.
- iii) Ability to process Large Data Volumes - can handle massive datasets that humans can't easily analyze.
- iv) Personalization - Enables ~~to~~ tailored recommendations, offers, and user experiences.
- v) Early Detection - Identifies patterns or anomalies (e.g. fraud, equipment failure) before issues escalate.

Disadvantages

- i) High Data dependency - performance relies heavily on the quality and quantity of data.
- ii) complexity - ML models often require advanced expertise.
- iii) computational cost - some models demand high processing power and storage.
- iv) potential Bias - Biased training data can lead to unfair or inaccurate predictions.
- v) Interpretability issues - many ML models are "black boxes" that are hard to explain.

Limitations

- i) Not a one-size-fits-all: models trained for one task usually can't be directly applied to another.
- ii) Struggles with unseen scenarios - Poor at predicting events that differ greatly from training data.
- iii) Needs ongoing maintenance - models require retraining to stay relevant as data changes.
- iv) Ethical & legal concerns - privacy issues, misuse risks, and compliance challenges can arise.
- v) Cannot Replace Human Judgement completely - Lacks common sense and contextual understanding.

 14/8/25

Assignment - 3

Date: SPS / /

Page No.

Name: Kaushal Kumar Ragh

Branch: B-Tech CSE with AI & ML

PRN: 240205241023

Semester: III

Submission date: 02/09/2025

Short answer questions:

① What is the purpose of a machine learning model?

→ At its core, an ML model is a mathematical approximation of a real-world process. It learns patterns from historical data so it can make predictions on unseen data. The whole point is generalization, not just memorization.

A model is a function $f: x \rightarrow y$

Goal is to minimize a loss function $L(x, \hat{y})$.

Example: predicting housing prices using size, no. of bedrooms, and location.

② Define stratified k-fold cross-validation with an example?

→ Standard k-fold CV splits data into k equal folds, but if the dataset is imbalanced, folds may not represent the whole dataset. Stratified ensures the same class proportions across data. Dataset divided into k folds. Each fold keeps label proportions consistent with the full dataset.

Ex: In medical diagnosis (90% healthy, 10% sick), stratified or ensure each fold has the same 9:1 ratio, preventing biased performance estimation.

③ What is the time complexity of computing mean and variance?

→ Both computations are linear because they require looking at every data point. variance requires the mean first.

$$\text{Mean} = \mu = \frac{1}{n} \sum_{i=1}^n x_i \rightarrow O(n)$$

$$\text{Variance} = \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 \rightarrow O(n)$$

Ex: with 1M sample, both mean and variance scale linearly → fast enough for big data.

④ Explain the concept of one-hot encoding in feature engineering?

→ Converts categorical variables into binary vectors, preventing models from thinking categories have numeric order. If a categorical variable has k categories then it produces k binary features.

Ex: Car brand = {Toyota, BMW, Tesla} - "BMW" → [0, 1, 0],

original	one-hot encoded
Gender	Gender male female
male	male 1 0
female	female 0 1
male	male 1 0
female	female 1 0

⑤ What is the space complexity of a feature-engineered dataset?

→ Adding features increases dataset dimensionality. High dimensional data causes more memory / storage usage (curse of dimensionality).

$$\text{Storage} = O(n \times d)$$

Ex: one-hot encoding 1000 categories for 10,000 rows gives 10M values stored.

⑥ Define a test set with an example?

→ A separate dataset not touched during training or validation. It's the final exam for the model. Data split ex: Train 70%, Validation 15%, Test 15%.

Example:

In Netflix recommendations, test data is a hidden set of user viewing patterns - used only for final evaluation.

⑦ What is the purpose of data quality in machine learning?

→ A model is only as good as the data it learns from. Noisy, biased, or incomplete data makes the model unreliable. Noise inflates variance, bias reduces accuracy.

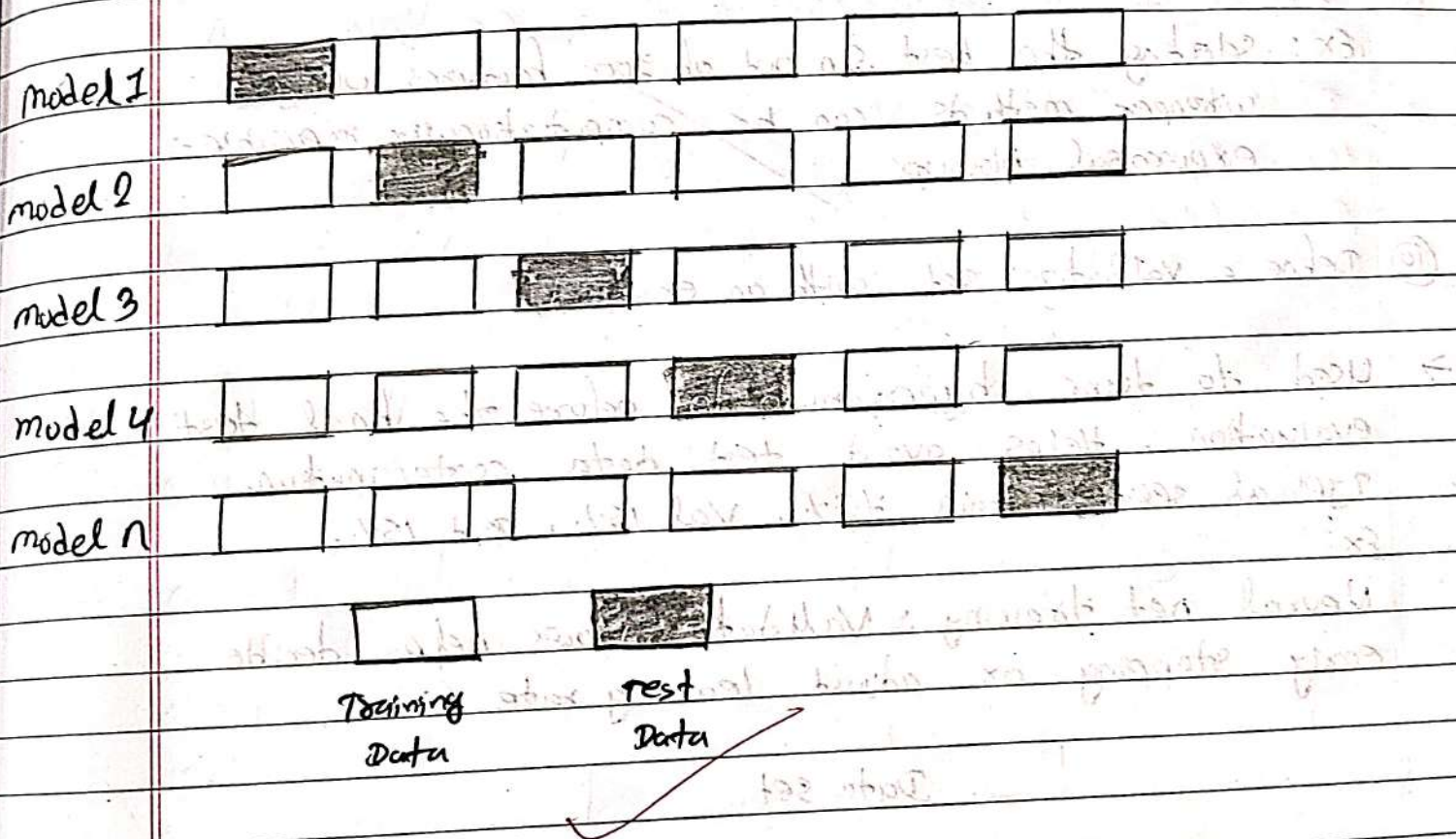
Ex: If a facial recognition dataset only contains lighter skin tones then there will be poor performance on darker skin: data quality failure.

② Explain the concept of leave-one-out cross-validation?

→ Extreme version of cv where each sample is used once as a test set. High accuracy, but very slow for large n . Here, $k=n$, complexity $\sim O(n^2)$.

Ex:

with 100 data point we train 100 models, each time leaving one sample out.



⑨ What is the time complexity of features selection?

→ Different methods:

- Filters: use statistics (correlation, chi-square).
- Wrapper: test subsets of features → slow.
- Embedded: selection inside training (e.g. Lasso).

Complexity: Filters: $O(nd)$

Wrapper: $O(2^d)$ worst-case

Embedded: $\sim O(nd)$

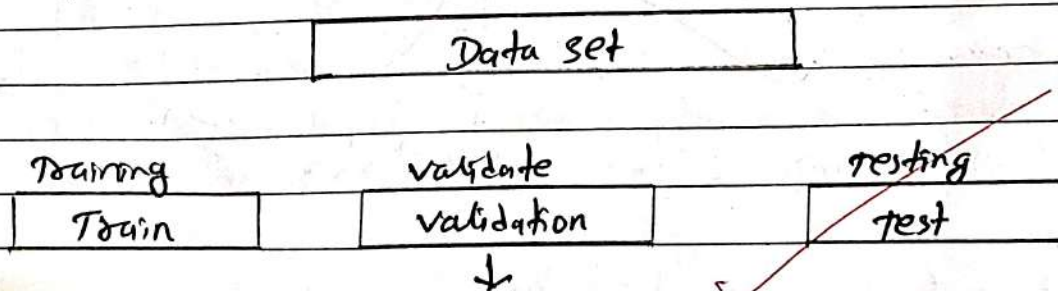
Ex: Selecting the best 20 out of 100 features using wrapper methods can be computationally impossible → exponential blowup.

⑩ Define a validation set with an example?

→ Used to tune hyperparameters before the final test evaluation. Helps avoid test data contamination.
Typical split: Train 70%, Val 15%, Test 15%.

Ex:

Neural net training: Validation loss helps decide early stopping or adjust learning rate.



tune Hyper parameters and select Best model

⑪ Define the K-Nearest Neighbour algorithm?

→ Non-parametric, lazy learner. Predictions are made by comparing a new point to stored training points.

For classification: $\hat{y} = \text{model}(y_i \text{ of } K \text{ nearest neighbours})$

ex:

Predicting if a movie will be liked by a user based on K nearest users with similar preferences.

⑫ What is the time complexity of training a linear regression model?

→ Theory: Two training approaches:

- Normal Equation: closed-form, exact but expensive
- Gradient Descent: iterative, scales better.

Math:

- Normal equation: $O(d^3)$ due to matrix inversion
- Gradient Descent: $O(ndk)$, with k = iterations.

ex: for small d (features), Normal eqn is fine for text data (thousand of features), GD is better.

⑬ Explain the concept of a confusion matrix?

→ Summarize classification outcomes into true/false/positive/negative. used to compute accuracy, precision, recall and $F1$:

• Accuracy = $(TP + TN) / \text{Total}$

• Precision = $TP / (TP + FP)$

• Recall = $TP / (TP + FN)$

Q14) What is the purpose of the SSE metric in regression?

→ SSE measures how badly predictions differ from true labels. Models aim to minimize SSE during training.

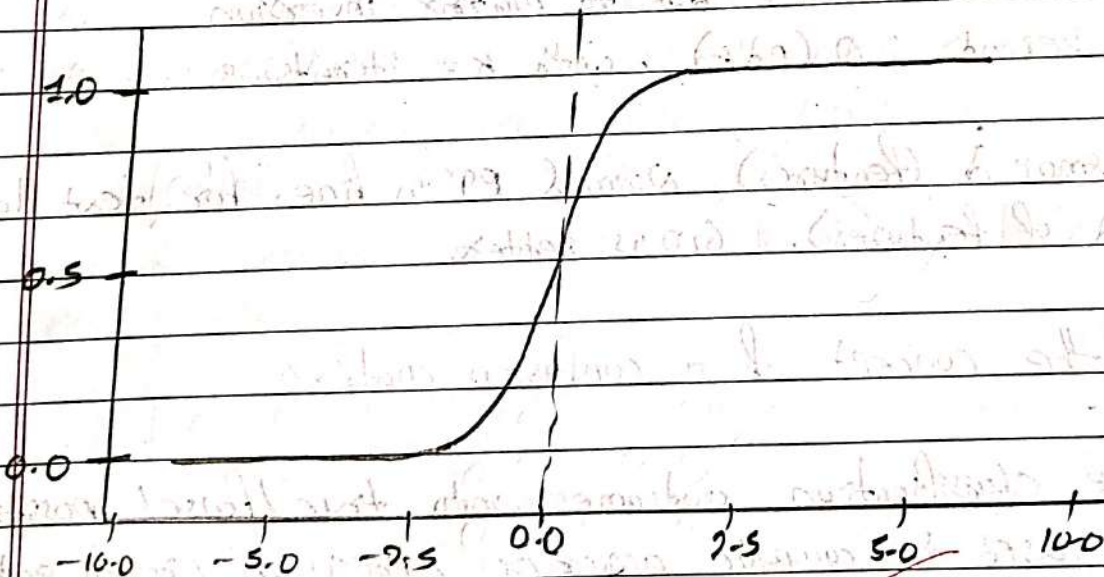
$$SSE = \sum (y_i - \hat{y}_i)^2$$

Ex: In predicting stock prices, smaller SSE means predictions are closer to reality.

Q15) Define logistic regression with an example?

→ Despite its name, it's a classification algorithm. Uses the sigmoid function to map values into probability.

$$\sigma(x) = \frac{1}{1 + e^{-x}} = \frac{1}{1 + e^{-(w \cdot x + b)}}$$



Ex: predicting if a student passes (1) or fails (0) based on the study hours and attendance.

Long Questions:

① Discuss the role of cross-validation in preventing overfitting.

→ Cross validation is a model validation technique that splits data into training and validation sets multiple times to evaluate model generalization.

A single train-test-split may give misleading result. The model may overfit (memorize) training data, failing on unseen data.

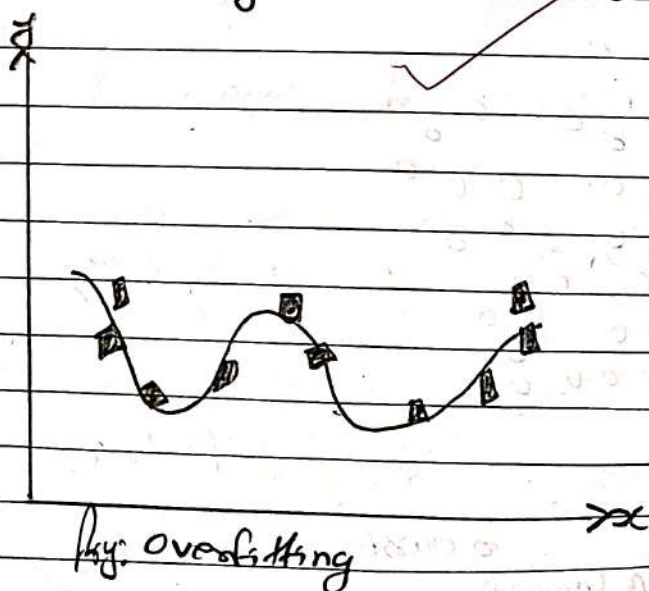
Formula (k-fold cv error) = $\frac{1}{k} \sum_{i=1}^k \text{Error}_i$

Types of CV:

- k-fold (most common)
- Leave-one-out (LOOCV)
- Stratified (for imbalanced datasets)

Ex: If you split data into 5 folds and the average accuracy is 92%, the model is likely generalizing well.

Conclusion: CV reduces overfitting by ensuring performance consistency across different subsets of data.



② Explain the implementation of k-nearest Neighbour for classification with an example?

→ KNN is a lazy, non-parametric algorithm that classifies a sample based on majority vote of its k nearest neighbours

Steps:

- 1) Choose k (odd number preferred)
- 2) Compute distance (Euclidean most common):

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

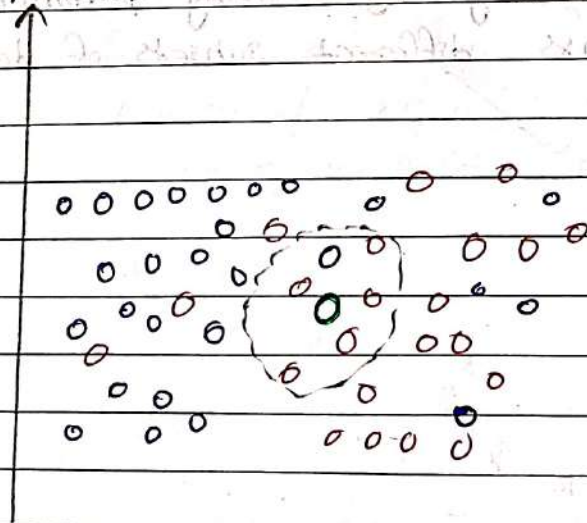
- 3) Select k nearest neighbours
- 4) Perform majority voting.

Ex: Classifying fruit based on weight & texture. If $k=3$.

• Distances → 2 apples, 1 orange → Predicted = Apple.

• Conclusion: KNN is simple, effective, but computationally expensive for large datasets.

Feature 2



Feature 1

○ class A

● = ?

● class B

class B (majority)

③ Discuss the time and space complexity of linear regression with examples?

→ Two implementations:

1) Normal Equation (closed form):

• Formula:

$$\theta = (X^T X)^{-1} X^T y$$

• Time: $O(n^2 m + n^3)$

• Space: $O(n^2)$

• works well for small n .

2) Gradient Descent:

• update rule:

$$\theta := \theta - \alpha \frac{1}{n} X^T (X\theta - y)$$

• Time: $O(mnk)$, where k = iterations.

• Space: $O(n)$

Example:

For 10,000 samples & 50 features → Gradient descent is faster than inverting a 50×50 matrix.

Normal equation is good for small features sets, while gradient descent is scalable.

④ Describe the role of evaluation metrics in supervised learning with examples?

→ Accuracy can be misleading (e.g. 95% accuracy but misses all fraud cases if only 5% fraud exists).

• Metrics & Formulas:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

• ROC-AUC = Area under True Positive Rate vs False Positive Rate curve.

Example: In medical diagnosis:

- Recall is crucial (catch all sick patients).
- Precision: avoid false alarms.

• Conclusion: Choice of metric depends on application.

⑤ Explain the implementation of a ~~confusion matrix~~ logistic regression for binary classification?

→ Logistic Regression models probability of output belonging to class 1.

Formula Sigmoid:

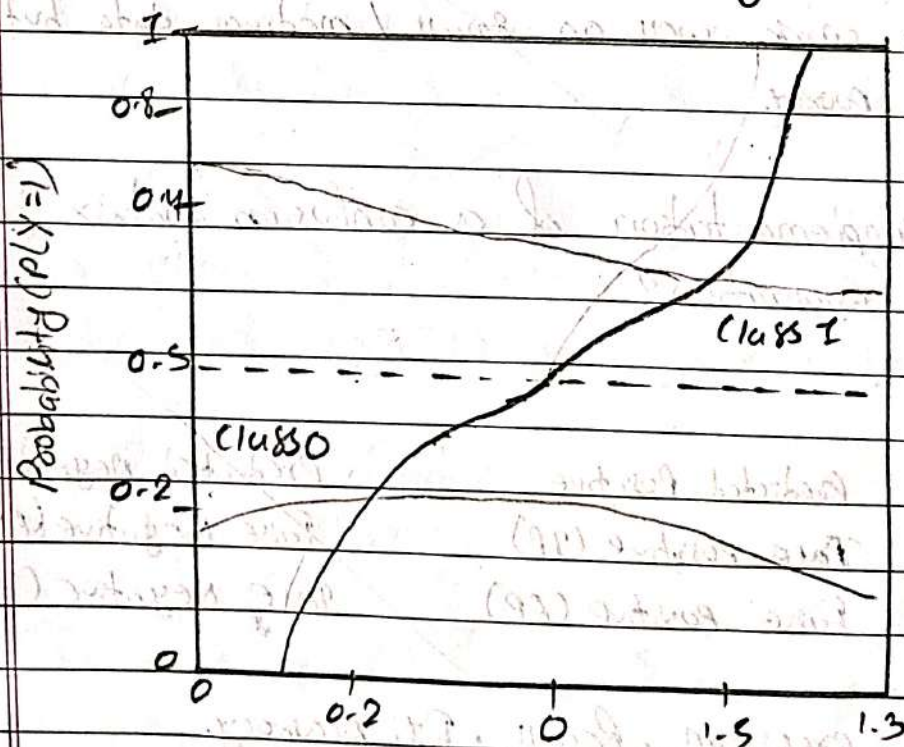
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$= \frac{1}{1 + e^{-(w \cdot x + b)}}$$

- Decision: if $\sigma > 0.5 \rightarrow \text{class 1}$ else class 0.
- Training: maximize likelihood using gradient descent.
- Example: Predicting if a student passes based on hours studied:

• $P = 0.82 \rightarrow$ student predicted to "Pass".

- Conclusion: Logistic regression is simple, interpretable, and widely used for binary classification.



Key: Linear combination of inputs (z)

⑥ Discuss the time and space complexity of Support Vector machines?

→ Training:

- Linear SVM: $O(mn)$ to $O(m^2n)$.
- Kernel SVM: $O(m^2n)$ to $O(m^3)$ (due to quadratic optimization)

Prediction:

- $O(s \cdot n)$, where s = support vectors.
- space: $O(m^2)$ storing kernel matrix
- Examples: for 50000 samples, training with kernel SVM is impractical, so linear / approximate methods are used
- Conclusion: SVMs work well on small / medium data but scale poorly.

⑦ Describe the implementation of a confusion matrix for classification evaluation?

→ structure:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

- Derived metrics: precision, Recall, F1, Accuracy.

- Example (Disease prediction):
 - TP = 50 (correctly predicted sick)
 - FP = 10 (healthy marked sick)
 - FN = 5 (sick missed)
 - TN = 100 (healthy predicted correctly)
- Conclusion: confusion matrix gives detailed error analysis beyond accuracy.

⑧ Explain the importance of precision, recall and F-score in classification?

-
- Precision: out of predicted positive, how many are correct.
 - Recall (Sensitivity): out of actual positives, how many are detected.
 - F1-score: Balance between precision & recall.

Formulas:

$$\text{Precision} = \frac{TP}{TP + FP}$$

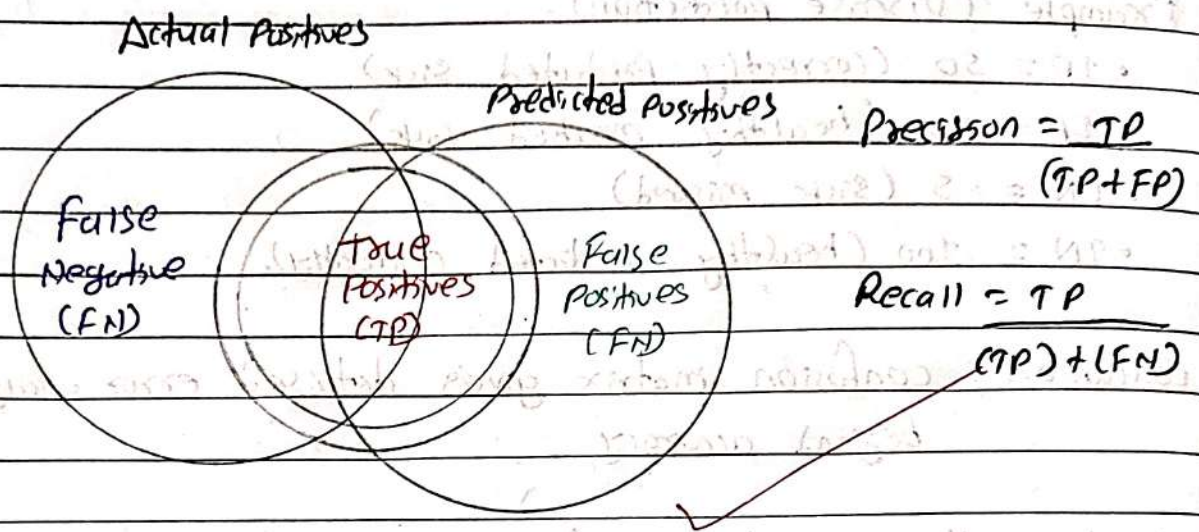
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2PR}{P+R}$$

Example:

- Fraud detection → Recall important
- medical test → Precision important
- Imbalanced dataset → F1-score is best.

Conclusion: Different applications prioritize different metrics.



Q) Discuss the applications of supervised learning in real-world scenarios?

→ Application are as:-

- i) Health care: Disease prediction (logistic regression).
- ii) Finance: credit scoring, fraud detection (svm, decision tree)
- iii) Retail: Product recommendation, demand prediction
- iv) Autonomous cars: Traffic sign recognition (CNNs)
- v) Example: Email spam filtering using Naive Bayes classifier

Supervised learning is used in nearly all industries where labeled data is available.

⑩ Discuss the role of cross-validation in preventing overfitting?

→ Cross-validation (CV) is a model validation technique that splits data into training and validation sets multiple times to evaluate model generalization.

A single train-test split may give misleading results. The model may overfit (memorize) training data, failing on unseen data.

$$k\text{-fold CV error} = \frac{1}{k} \sum_{i=1}^k \text{Error}_i$$

Types:

- k-fold CV
- Stratified k-fold (for imbalanced data)
- leave-one-out CV

Example: Fraud detection → Stratified CV ensures equal ratio of fraud / non-fraud in each fold.

Cross validation ensures stability, prevents overfitting and provides reliable model.

Signature

Assignment:- 4

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Branch: B-Tech CSE AI & ML

Semester: III

Subject: Machine Learning.

1) What is the time complexity of predicting with K-Nearest Neighbor?

→ The time complexity of predicting with KNN is $O(n \times d)$, where n is the number of training samples and d is the number of features. This is because, for each prediction, KNN calculates the distance between the test point and every training point. If sorting is required to find the top k nearest neighbours, it can go up to $O(n \times d + n \log k)$.

2) Explain the concept of precision in classification.

→ Precision measures how accurate the positive predictions of a classifier are. It's calculated as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where,

TP = True Positives

FP = False Positives

In simple terms, it tells us: "of all the instances predicted as positive, how many were actually positive?"
High precision means fewer false positives.

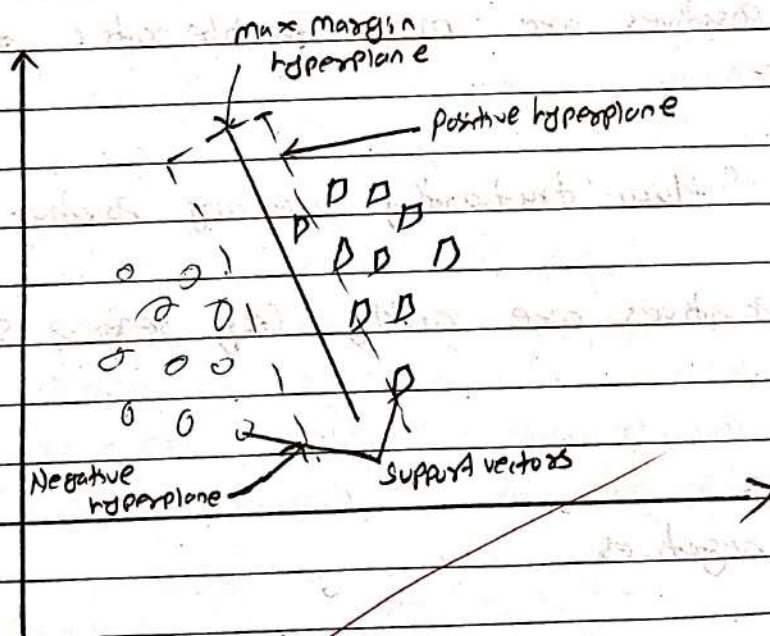
3) What is the space complexity of a linear regression model?

→ A linear regression model stores one weight for each feature and a single bias term. Therefore, its space complexity is $O(d)$, where d is the number of features. Once trained, it doesn't need to retain the training data, keeping memory usage minimal.

4) Define the Support Vector Machine (SVM) Algorithm.

→ Support vector machine (SVM) is a supervised learning algorithm that finds the optimal hyperplane separating data points of different classes while maximizing the margin between them.

- Support vector machine is used for classification and regression. It works by finding decision planes that separate data into different classes.



⑤ What is the time complexity of training a SVM?

→ Training a SVM involves solving a quadratic optimization problem, making it computationally heavy. The time complexity typically ranges between $O(n^2 \times d)$ and $O(n^3)$, depending on the dataset on the dataset and kernel used. Linear SVM or approximate solvers can reduce this complexity for large datasets.

⑥ Explain the concept of Precision in classification.

→ Precision is the proportion of true positive predictions among all positive predictions made by the model.

$$\text{Precision} = \frac{TP}{TP + FP}$$

It focuses on correctness rather than completeness - useful when false positives are more problematic than false negative.

Interpretation: "How trustworthy are my positive predictions?"

Good for:

- when false positives are costly (e.g. sending spam mails to legit users).

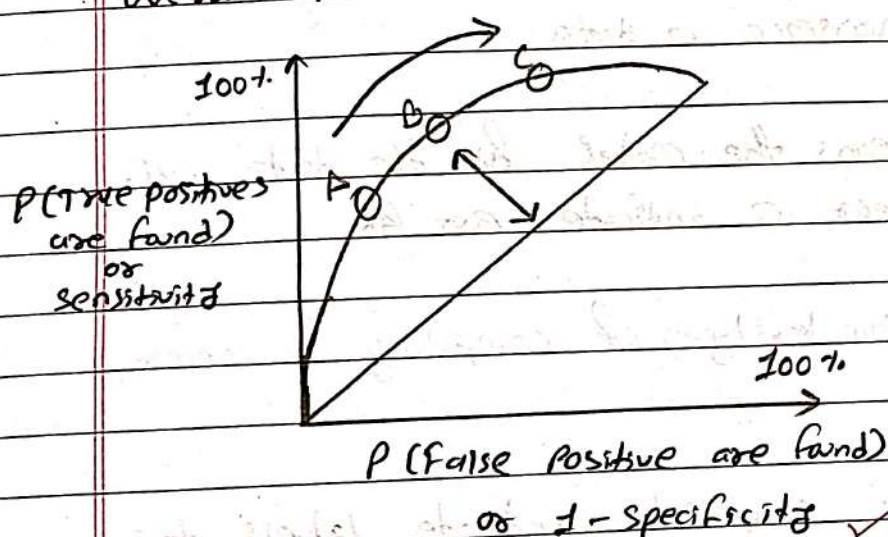
Weakness:

- Ignores false negatives

⑦ What is the purpose of the Roc curve in classification?

→ The Roc (Receiver operating characteristic) curve shows how the True Positive Rate (TPR) and False Positive Rate (FPR) change across different classification thresholds. It helps evaluate a model's ability to distinguish between classes.

The AUC (Area under curve) represents the model's overall performance - the closer to 1, the better.



⑧ What is the space complexity of a K-Nearest Neighbour model?

→ KNN is a lazy learner, meaning it stores the entire training dataset for use during prediction. Thus, its space complexity is $O(n \times d)$, where n is the number of samples and d is the number of features. Memory usage scales directly with dataset size.

⑨ Define the R^2 metric in regression

→ The R^2 (coefficient of determination) measures how well the regression model explains the variability of the target variable.

$$R^2 = 1 - \frac{SSE}{SST}$$

Where SSE = sum of squared errors

SST = total variance in data

An R^2 close to 1 means the model fits the data well, while ~~low~~ values near 0 indicate poor fit.

⑩ What is the time complexity of computing a confusion matrix?

→ The confusion matrix compares predicted labels to actual labels for n samples, so the time complexity is $O(n)$. It's a simple counting process that classifies each prediction into one of the four outcomes: TP, FP, TN, FN.

⑪ Explain the concept of recall in classification.

→ Recall (or sensitivity) measures how well a model identifies actual positives.

$$\text{Recall} = \frac{TP}{TP + FN}$$

It answers: "Of all true positives, how many did the model correctly detect?"

High recall means fewer false negatives - ideal for tasks like medical diagnosis.

(12) What is the space complexity of a support vector machine model?

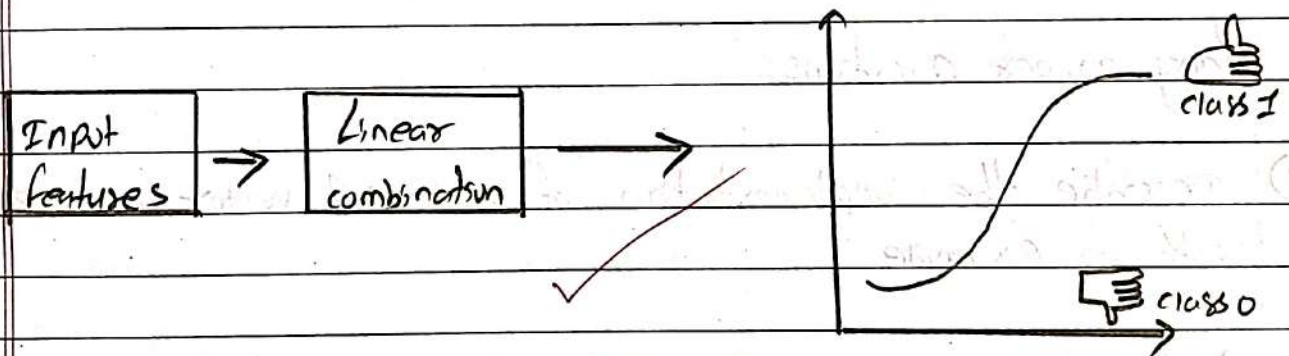
→ The space complexity of mainly depends on the number of support vectors after training. It's approximately $O(s \times d)$, where s is the number of support vectors and d is the number of features. For non-linear kernels, storing the kernel matrix can rise to $O(n^2)$.

(13) Define a classification task with an example.

→ A classification task assigns data samples into predefined categories based on their features.

Example:

Predicting whether an email is spam or not spam using features like word frequency, sender and subject line.



(14) What is the time complexity of evaluating a regression model?

→ Evaluating a regression model (like computing RMSE, MAE or R^2) requires comparing predicted and actual values for all samples.

Hence,

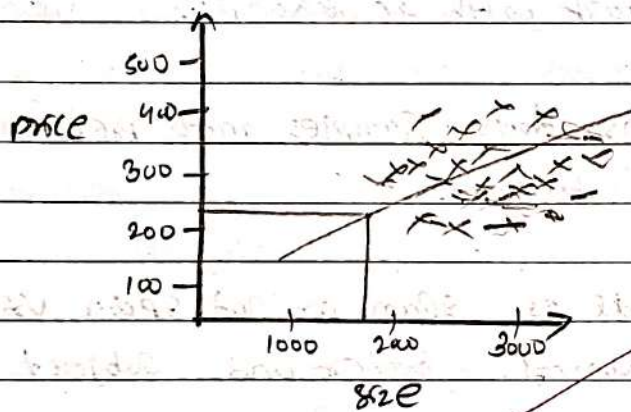
the time complexity is $O(n)$, where n is the number of test samples.

⑤ Explain the concept of a regression task with an example

→ A regression task predicts a continuous numerical value rather than a class label.

Example: Predicting house prices based on features like size, location, and ~~human~~ number of rooms.

The goal is to minimize the difference between predicted and actual values.



Long answer questions:

1) Describe the implementation of a support vector machine with an example.

→ A SVM is a supervised machine learning algorithm primarily used for classification and regression tasks. It works by finding an optimal hyperplane that best separates data points of different classes with the maximum margin (distance between the hyperplane and the nearest data points).

Implementation steps:

1) Data Preparation:

Load and clean the dataset, handle missing values, and normalize features since SVMs are sensitive to feature scaling.

2) Splitting the data:

Divide the dataset into training and testing sets.

3) Model Training:

Use an SVM classifier (like `SVC` from `sklearn`) and choose a kernel - eg. Linear, polynomial, or RBF depending on data patterns.

4) Prediction & Evaluation:

Example :

```
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
```

```
X, y = datasets.load_iris(return_X_y=True)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y,
                                                    test_size=0.2)
```

```
model = SVC(kernel='linear')
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, predictions))
```

Like classifying emails as spam or not spam based on textual features.

② Explain the role of the ROC curve in evaluating classification models.

→ The ROC (Receiver operating characteristics) curve is a graphical representation of a classification model's performance across different threshold settings. It plots:

- True Positives Rate (TPR) on the y-axis
- False Positives Rate (FPR) on the x-axis

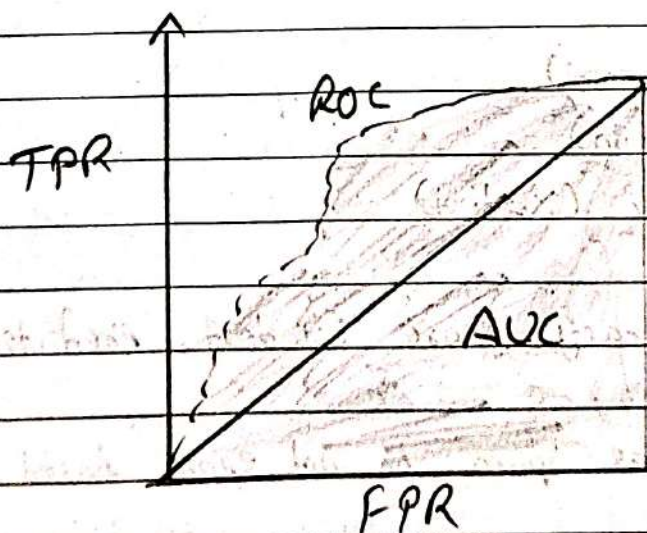
Each point on the curve represent a different threshold for deciding whether a sample belongs to the positive class;

Purpose:

- Shows the trade-off betwⁿ sensitivity (recall) and specificity
- Helps compare multiple models.
- A model's that perform well has a curve closer to top-left corner.

AUC (Area under curve):

- The AUC value summarizes the overall performance
- $AUC = 1 \rightarrow$ perfect classifier
- $AUC = 0.5 \rightarrow$ Random guess
- Higher AUC means better model performance



③ Discuss the time and space complexity of k-Nearest Neighbour (KNN).

→ Time complexity:

- Training: $O(1)$ → KNN is a lazy learner; it doesn't train explicitly.
- Prediction: $O(nd)$ → Each test sample requires distance calculation with all n training samples having d features.
Sorting neighbours adds $O(n \log k)$ overhead.

Space complexity:

- $O(nd)$ since KNN stores the entire training dataset for predictions.

Trade off: High memory and slower predictions, but easy to implement and highly interpretable.

④ Describe the implementation of grid search for hyperparameter tuning.

→ Grid search is a brute-force method used to find the best hyperparameter combination for a model.

Steps:

1) Define the model:

Choose the algorithm (e.g. SVM, Random Forest)

2) Create a parameter grid:

List the possible values for each hyperparameter

3) Cross-validation

Use k-fold CV to evaluate model performance for every parameter combination.

4) Select Best Parameters:

The combination with the highest cross-validation accuracy is chosen.

Example:

```
from sklearn.model_selection import GridSearchCV
from sklearn.svm import SVC
```

```
Param_grid = {'C': [0.1, 1, 10], 'kernel': ['linear', 'rbf']}
grid = GridSearchCV(SVC(), Param_grid, cv=5)
grid.fit(X_train, y_train)
```

Print ("Best Parameters:", grid.best_params_)

③ Explain the role of MME metric in regression evaluation

⇒ MME refers to mean magnitude of error or mean model error like similar to MAE / MSE

The MME (mean model error) is used to measure how far the predicted value are from the actual ones in regression tasks. It captures the average prediction error magnitude

$$MME = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- y_i = actual value, $\hat{y}_i$ = predicted value

Interpretation:

- Lower MME means better model performance
- It is simple, interpretable, and less sensitive to outliers than MSE

⑥ Explain unsupervised learning. Discuss clustering and dimensionality reduction techniques.

→ Unsupervised learning deals with unlabeled data, the algorithm identifies hidden patterns or structure without explicit supervision.

i) Clustering:

Grouping similar data points together based on feature similarity

- K-means clustering: Divides data into k clusters by minimizing intra-cluster variance.

- Hierarchical clustering: Builds nested clusters in a tree structure

- DBSCAN: Detects clusters based on density, great for irregular shapes.

ii) Dimensionality Reduction:

Reducing the no of input features while retaining key info.

- PCA (Principal Component Analysis): Projects data into fewer orthogonal dimensions.

- t-SNE / UMAP: Used for visualization of high-dimensional data

- Autoencoders: Neural Network that compresses and reconstruct data

⑦ Explain the concept of Reinforcement Learning with a suitable real-life example.

→ Reinforcement learning is a type of learning where an agent interacts with an environment, learns through trial and error, and gets rewards or penalties based on its actions.

key components

- Agent: learner or decision-maker
- Environment: everything the agent interact with
- Action: what the agent can do
- State: The current situation of the environment
- Reward: feedback signal from the environment

example: self-driving cars learn to navigate roads by getting positive rewards for staying in the lane and reaching destinations safely, and negative rewards for collisions.

⑧ Describe any three machine learning applications in business and industry.

→ 1) Fraud detection (Finance):

ML models like Random forests detect abnormal transaction patterns.

2) Customer segmentation (Marketing):

- Clustering group customers based on purchase behaviour for promotions.

3) Predictive maintenance (Manufacturing):

- Regression and time-series models predict equipment failures, reducing downtime and maintenance cost.

Q) Discuss the importance of data preprocessing in the machine learning process.

→ Data preprocessing transforms raw data into a clean and usable format for ML models.

It's crucial because poor-quality data = poor model performance

Key Steps:

- Data Cleaning: Handle missing values, outliers, and inconsistencies.
- Data Transformation: Normalize or standardize features.
- Encoding: Convert categorical variables into numerical form.
- Feature Selection: Remove redundant or irrelevant features.

Importance:

- Improves accuracy and model stability
- Reduces training time
- Enhances generalization to unseen data.

without preprocessing, even the best algorithm can fail

Q) Explain the advantages, disadvantages and limitations of machine learning?

Advantages

- Automation: Learns patterns and automates complex decision-making
- Continuous Improvement: Improves with more data and retraining
- Broad Applicability: used across healthcare, finance, education etc.
- Handle complex data: can manage high-dimensional and non-linear data.

Disadvantages:

- Data Dependency: Needs large, quality datasets to perform well.
- High computation cost: Training advanced models requires significant resources.
- Interpretability: Some models (like deep learning) acts as "black boxes".

Limitations:

- Bias & ethics: models can inherit bias from data.
- Overfitting: models can perform well on training data but fails on new data.
- Security risks: vulnerable to data poisoning and adversarial attacks.

[Signature]

Assignment:- 5.

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Semester: III

Subject: Machine Learning

① What is the purpose of the MME metric in regression?

→ The Mean Model (MME) is used to evaluate the average difference predicted and actual values in regression problems.

It tells how close the predictions are to the real outputs on average:

$$MME = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

A smaller MME means better performance. It's a simple and interpretable metric that, unlike Mean squared error, doesn't exaggerate large errors - making it more robust against outliers.

② Explain the concept of hyperparameter tuning in SVM.

→ In a SVM (Support Vector machine), hyperparameters determine how the model separates data. They aren't learned automatically but must be set manually or tuned.

Key hyperparameters include:

- C (Regularization): controls the trade-off between maximizing margin and minimizing misclassification.

- **kernel type:** Determines the function used to transform data (linear, polynomial, RBF, etc).
- **Gamma:** Influences the curvature of the decision boundary for non-linear kernels.

Hyperparameters tuning uses techniques like Grid search or Random search to test different combinations and finds the set that provides the best accuracy or F1-score on validation data.

③ What is the time complexity of predicting with a logistic regression model?

→ Prediction in logistic regression requires calculating a weighted sum of features (dot product) and applying the sigmoid function.

- For one sample $\rightarrow O(d)$
- For n samples $\rightarrow O(n \times d)$

It's fast because both the dot product and sigmoid are linear-time operations.

④ Define a soft-margin support vector machine

→ A soft margin SVM allows training points to be misclassified so the model can handle noisy or overlapping data.

It introduces slack variable (ξ_i) and a penalty parameter (C):

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

A small C allows ^{more} margin violations (more generalization), while a large C enforces stricter classification with a narrower margin.

This flexibility helps in real-world data that's rarely perfectly separable.

⑤ What is the space complexity of a confusion matrix?

→ A confusion matrix for a classification task with k classes is a $k \times k$ table.

Each cell stores the count of predictions belonging to a particular actual - predicted combination.

Thus, the space complexity = $O(k^2)$.

For binary classification ($k=2$), it's fixed 2×2 matrix - constant space ($O(1)$).

⑥ Explain the concept of the AUC metric in classification.

→ AUC (Area under ~~ROC~~ ROC curves) measures how well a model distinguishes between positive and negative classes. It represents the probability that the model ranks a randomly chosen positive instance higher than a negative one.

- AUC = 1: perfect classifier
- AUC = 0.5: Random prediction
- AUC < 0.5: poor model

A higher AUC value means the model is better at separating classes across all threshold levels.

⑦ What is the time complexity of computing recall?

→ Recall measures the proportion of actual positives correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

To compute recall, the algorithm scans all predictions once to count true positives (TP) and false negatives (FN).

Therefore, the time complexity = $O(n)$, where n is the number of samples.

⑧ What is the space complexity of a logistic regression model?

→ A logistic regression model stores:

- weights (w): one per feature $\rightarrow O(d)$
- Bias (b): single value $\rightarrow O(1)$

Here,
total space complexity = $O(d)$.

This makes logistic regression memory-efficient and scalable to high-dimensional data.

⑨ Explain the concept of grid search in supervised learning.

→ Grid Search is a systematic hyperparameter optimization method used in supervised learning.

It builds a "grid" of parameter combinations, trains models for each, and selects the one with the best cross-validation score.

example:

For SVM:

params = {'C': [0, 1, 2, 10], 'kernel': ['linear', 'rbf']}

Each combination of C and kernel is tested. The one yielding the best accuracy or F1-score is chosen. While effective, Grid Search can be computationally heavy for large parameter spaces.

⑩ Define a multi-class classification task with an example.

→ A multi-class classification problem involves predicting one label from three or more classes.

Example:

Handwritten digit recognition (digit 0-9) using the MNIST dataset, each image belongs to one of ten classes.

Common algorithms include Decision Tree, Random Forests, Softmax Regression, and multi-class SVMs (one-vs-rest or one-vs-one approaches).

⑪ What is the time complexity of training a k-Nearest Neighbour model?

→ KNN doesn't have a true "training phase" it simply stores the dataset.

- Training time complexity : $O(1)$
- Prediction time complexity : $O(n \times d)$, since it computes distances between the test sample and all n training samples (each with d features).

This makes KNN slow during prediction but extremely fast to set up.

⑫ Define the precision-recall curve with an example

→ The precision-recall curve shows the how precision and recall vary as the decision threshold changes.

- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$

A good model has a curve close to the top-right corner (high precision and recall).

Example:

in fraud detection, lowering the threshold may increase recall (catch more frauds) but reduce precision (more false alarms).

(13) What is the space complexity of storing model predictions?

→ If a model produces predictions for n samples, and each prediction is a single scalar value, the space complexity is $O(n)$.

For multi-class problems with k classes (probabilities per class), the complexity becomes $O(n \times k)$.

(14) Explain the concept of a binary classification task.

→ Binary classification means predicting one of two possible categories.

Examples:

- Email spam detection (spam / not spam)
- Medical test result (Positive / Negative)
- Credit card fraud detection (Fraud / Normal)

It's the simplest classification type and evaluated using metrics like Accuracy, Precision, Recall, F1 and AUC.

Like two classes (0 or 1)

Binomial

(15) What is the time complexity of computing the R^2 metric?

→ The R^2 (coefficient of determination) measures how well a regression model explains the variance in target

$$R^2 = 1 - \frac{SS_{res}(E)}{SS_{tot}(T)}$$

It requires computing predicted and actual differences once across all n samples, making the time complexity $= O(n)$.

It's efficient and widely used for regression model evaluation

Long answer questions.

(1) Describe the applications of regression model in real-world scenarios.

→ Regression models are used to predict continuous values and understand relationships between variables

Common applications:

- Finance: Predicting stock prices, credit risk or loan defaults.
- Real Estate: Estimating house price based on features like area, location, and number of rooms.
- Healthcare: Forecasting disease progression or predicting patient recovery times.
- Marketing: Estimating sales or advertising impact using historical data.
- Environment: Predicting pollution levels or weather conditions.

② Explain the implementation of a soft-margin SVM with an example.

→ A soft-margin SVM allows some misclassified data points to handle noisy or overlapping datasets. It introduces slack variable (ξ) and a penalty parameter (C) to balance between maximizing the margin and minimizing misclassification.

Steps:

- 1) Normalize and prepare labeled data
- 2) Define the objective

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum \xi_i$$
- 3) Train using a kernel (like linear or RBF)
- 4) Predict new samples based on the decision boundary.

Example (Python):

```
from sklearn import svm
model = svm.SVC(kernel='linear', C=1.0)
model.fit(X_train, y_train)
```

Here,

$C=1.0$ controls the trade off - smaller C gives a wider margin, larger C fits data more tightly.

③ Discuss the role of the AUC metric in evaluating classifier performance

- AUC (Area under Roc curve) measures how well a classifier separates classes. It represents the probability that a randomly chosen positive sample is ranked higher than a negative one.
- AUC = 1: perfect model
 - AUC = 0.5: Random guessing

It's especially useful for imbalanced datasets, as it evaluates performance across all thresholds rather than relying on a single accuracy value.

Example (Python):

```
from sklearn.metrics import roc_auc_score
auc = roc_auc_score(y_test, model.predict_proba(x_test)[:,1])
```

A higher AUC means better overall classification performance.

④ Explain the implementation of k-means clustering with a real-world example.

- k-means is an unsupervised learning algorithm used to group similar data point into k clusters.

Steps:

- 1) Choose the number of clusters (k)
- 2) Randomly initialize k centroids
- 3) Assign each data point to the nearest centroid.

- 4) update centroids as the mean of assigned points
- 5) Repeat until centroids stabilize

Example:

Customer segmentation - grouping customers by purchase behaviour to target marketing strategies.

Code:

from sklearn.cluster import KMeans

kmeans = KMeans (n_clusters=3, random_state=0)

kmeans.fit(x)

labels = kmeans.labels

⑤ What is the purpose of the MME metric in regression?

→ MME (Mean Model Error) measures the average absolute difference between predicted and actual values:

$$MME = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

It gives a direct sense of prediction accuracy - lower MME indicates better model performance, compared to MSE, it's less sensitive to outliers and easier to interpret.

Example:

If an energy consumption model predicts 102 units when the actual usage is 100, the error is 2. Averaging such errors gives MME - a clear estimate of model accuracy.

15/11/25